

Digital Image Processing Terminal Exam — Complete Solutions

CSC / CEN — Digital Image Processing

Spring 2024 · Final Terminal Paper

Step-by-step worked solutions to all six questions of the Spring 2024 terminal examination, including histogram equalization, adaptive median filtering, color transformations, morphological dilation, edge profiles, and GLCM analysis.

Corrected Edition — Q5A edge profile diagrams fixed.

Q1
Histogram EQ
5 marks

Q2
Adaptive Median
8 marks

Q3
Color Processing
7 marks

Q4
Morphological Dilation
4 marks

Q5
Edge Profiles & DoG
16 marks

Q6
Chain Codes & GLCM
10 marks

Histogram Equalization

PROBLEM STATEMENT

The probabilities of individual gray levels for a 3-bit coded image are given in the table below. Apply histogram equalization and compute the new pixel intensities after equalization. Show the original and equalized histograms, as well as the transformation function.

Intensity r_k	Probability $P_r(r_k)$
$r_0 = 0$	0.20
$r_1 = 1$	0.30
$r_2 = 2$	0.25
$r_3 = 3$	0.15
$r_4 = 4$	0.10
$r_5 = 5$	0.00
$r_6 = 6$	0.00
$r_7 = 7$	0.00

SETUP

3-bit image: $L = 2^3 = 8$ gray levels

Maximum intensity: $L - 1 = 7$

Equalization formula:

$$s_k = \text{round}(7 \times \text{CDF}_k)$$

where $\text{CDF}_k = \sum_{j=0}^k P_r(r_j)$

Step 1 — CDF Computation & Transformation Mapping

k	r_k	$P_r(r_k)$	CDF_k	$7 \times \text{CDF}_k$	$s_k = \text{round}(7 \times \text{CDF})$
0	0	0.20	0.20	1.40	1
1	1	0.30	0.50	3.50	4
2	2	0.25	0.75	5.25	5
3	3	0.15	0.90	6.30	6
4	4	0.10	1.00	7.00	7
5	5	0.00	1.00	7.00	7
6	6	0.00	1.00	7.00	7
7	7	0.00	1.00	7.00	7

Step 2 — Transformation Mapping Summary

$r_0=0 \rightarrow s_0=1$

$r_1=1 \rightarrow s_1=4$

$r_2=2 \rightarrow s_2=5$

$r_3=3 \rightarrow s_3=6$

$r_4=4 \rightarrow s_4=7$

$r_5=5 \rightarrow s_5=7$

$r_6=6 \rightarrow s_6=7$

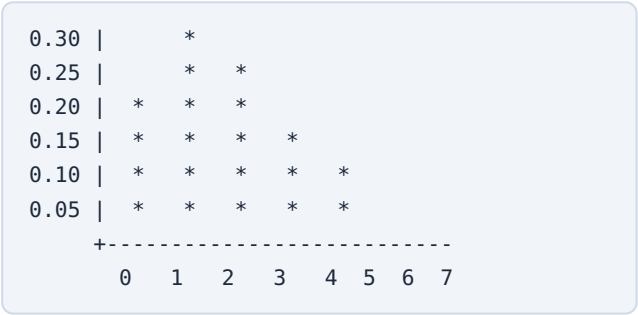
$r_7=7 \rightarrow s_7=7$

Step 3 — Equalized Output Probabilities $P_s(s_i)$

Output Level s_i	Source	$P_s(s_i)$
0	None	0.00
1	$\leftarrow P_r(0) = 0.20$	0.20
2	None	0.00
3	None	0.00
4	$\leftarrow P_r(1) = 0.30$	0.30
5	$\leftarrow P_r(2) = 0.25$	0.25
6	$\leftarrow P_r(3) = 0.15$	0.15
7	$\leftarrow P_r(4)+P_r(5)+P_r(6)+P_r(7)$	0.10

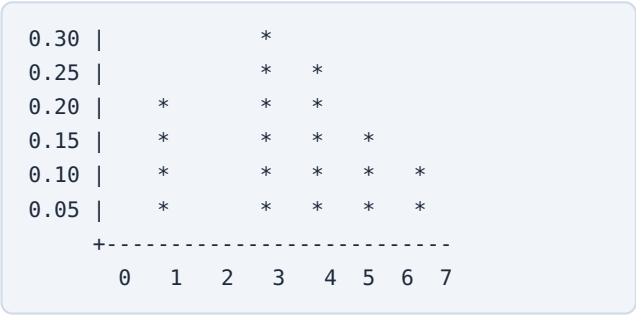
Step 4 — Histograms (ASCII representation)

Original Histogram $P_r(r_k)$



Levels 0-4 only; heavily left-skewed

Equalized Histogram $P_s(s_i)$



Spread across range [1,7]; levels 2,3 empty

TRANSFORMATION FUNCTION BEHAVIOUR

The mapping $r_k \rightarrow s_k$ is a staircase function that is **monotonically non-decreasing**. Levels 5, 6, 7 all map to 7 because the original image contained no pixels at those intensities ($P_r = 0$). This is normal for discrete histogram equalization.

Sorted:

Z_min = 0 Z_med = 3 Z_max = 8

LEVEL A

$$A1 = 3 - 0 = 3 \text{ (> 0 OK)}$$

$$A2 = 3 - 8 = -5 \text{ (< 0 OK)}$$

→ Proceed to Level B.

LEVEL B

$$B1 = 0 - 0 = 0 \text{ (NOT > 0)}$$

→ Z_{xy} IS an impulse (pepper).

$$\text{Output} = Z_{\text{med}} = 3$$

Pixel 3 — $Z_{xy} = 1$ at position (Row 3, Col 2)



LEVEL A

$$A1 = 3 - 0 = 3 \text{ (> 0 OK)}$$

$$A2 = 3 - 8 = -5 \text{ (< 0 OK)}$$

→ Proceed to Level B.

LEVEL B

$$B1 = 1 - 0 = 1 \text{ (> 0 OK)}$$

$$B2 = 1 - 8 = -7 \text{ (< 0 OK)}$$

→ Z_{xy} is NOT impulse.

$$\text{Output} = Z_{xy} = 1$$

Pixel 4 — $Z_{xy} = 8$ at position (Row 3, Col 3)



LEVEL A

$$A1 = 3 - 0 = 3 \text{ (> 0 OK)}$$

$$A2 = 3 - 8 = -5 \text{ (< 0 OK)}$$

→ Proceed to Level B.

LEVEL B

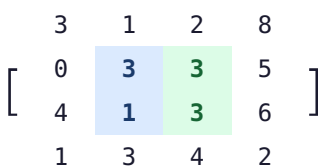
$$B1 = 8 - 0 = 8 \text{ (> 0 OK)}$$

$$B2 = 8 - 8 = 0 \text{ (NOT < 0)}$$

→ Z_{xy} IS impulse (salt).

$$\text{Output} = Z_{\text{med}} = 3$$

Final Output Matrix



Output Image G

SUMMARY OF CHANGES

- Pixel (2,2): 3 → 3 (unchanged)
- Pixel (2,3): 0 → 3 (pepper noise removed)
- Pixel (3,2): 1 → 1 (unchanged)
- Pixel (3,3): 8 → 3 (salt noise removed)

Blue = unchanged. Green = corrected (impulse removed).

Color Image Processing

Part A — Middle Column Prediction & Inverse Transformation

At position $N/2$ (the middle column), the three channel profiles give:

CHANNEL VALUES AT $x = N/2$

- **Green (G):** Linear ramp 1.0 to 0.0 $\Rightarrow G = 0.5$
- **Red (R):** Triangle peaking at $N/2 \Rightarrow R = 1.0$
- **Blue (B):** Linear ramp 0.0 to 1.0 $\Rightarrow B = 0.5$

Combined vector: $(R, G, B) = (1.0, 0.5, 0.5)$

COLOUR DECOMPOSITION

$(1.0, 0.5, 0.5) = \text{gray}(0.5, 0.5, 0.5) + \text{pure red}(0.5, 0, 0)$

This is **Pink** (desaturated red / light red).



Pink = (R:1.0, G:0.5, B:0.5)

AFTER NEGATIVE (INVERSE) TRANSFORMATION

$(1-R, 1-G, 1-B) = (0.0, 0.5, 0.5)$

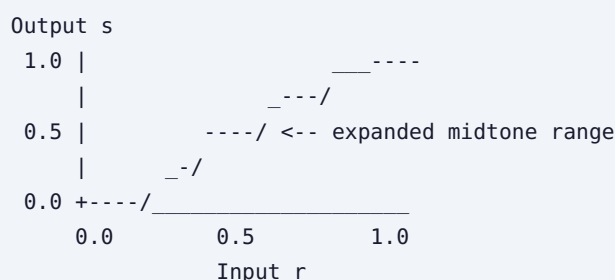
$G = B = 0.5$ with $R = 0 \Rightarrow$ **Mid-intensity Cyan**



Cyan = (R:0.0, G:0.5, B:0.5)

Part B — Transformation Curve to Correct Low Contrast

To expand low contrast (stretch midtones while compressing highlights and shadows), use an **S-shaped (sigmoidal) transformation curve**:



An S-curve maps dark inputs to even darker outputs (compressed shadows), middle inputs to a wide spread (expanded midtones), and bright inputs to near-maximum outputs (compressed highlights). This increases apparent contrast in the mid-range.

Part C — Monitor vs. Printer Yellow Correction

If a balanced screen image prints as **yellowish**:

ADDITIVE (RGB) CORRECTION

- Yellow = Red + Green (additive)
- **Decrease the Red** channel
- **Decrease the Green** channel

SUBTRACTIVE (CMYK) CORRECTION

- Yellow ink (Y) is directly responsible
- **Reduce Yellow (Y) ink**
- Or **increase Cyan (C) + Magenta (M)** to produce Blue, neutralising excess Yellow

Binary Morphological Dilation

PROBLEM STATEMENT

Perform dilation of image A with a symmetric 3x3 cross structuring element B.

$A = \{ (2,2), (4,2), (3,6), (2,9), (3,9) \}$ $B = \text{symmetric } 3 \times 3 \text{ cross (origin at centre)}$

DILATION RULE

For each active coordinate in A, place the cross SE centred on that point and mark all cells it covers. The cross SE adds offsets: $\{(0,0), (-1,0), (+1,0), (0,-1), (0,+1)\}$ to each seed coordinate.

Expansion per Seed Coordinate

CONTRIBUTIONS OF EACH SEED

- **(2,2):** covers (1,2), (2,1), (2,2), (2,3), (3,2)
- **(4,2):** covers (3,2), (4,1), (4,2), (4,3), (5,2)
- **(3,6):** covers (2,6), (3,5), (3,6), (3,7), (4,6)
- **(2,9):** covers (1,9), (2,8), (2,9), (2,10), (3,9)
- **(3,9):** covers (2,9), (3,8), (3,9), (3,10), (4,9)

CROSS SE (B)

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & * & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Offsets: N, S, E, W + centre

Output Dilation Grid

Purple = original seed pixels. Blue = pixels added by dilation.

Row\Col	1	2	3	4	5	6	7	8	9	10
Row 1	.	+	.	.	.	.	.	.	+	.
Row 2	+	0	+	.	.	+	.	+	0	+
Row 3	.	+	.	.	+	0	+	+	0	+
Row 4	+	0	+	.	.	+	.	.	+	.
Row 5	.	+	.	.	.	.	.	.	.	.

0 = original seed | + = added by dilation | . = background

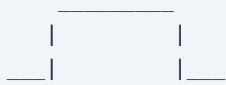
Edge Profiles, DoG Proof & Quad Tree

Part A — Gradient vs. Laplacian Edge Profiles

Plateau (Step Edge) — Complete Box Profile

Corrected: The intensity profile shows both rising AND falling edges (complete rectangular pulse). Gradient shows +spike (rising) and -spike (falling). Laplacian shows bipolar doublets at both transitions.

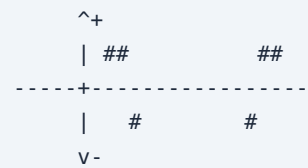
Intensity $f(x)$:



Gradient df/dx :



Laplacian d^2f/dx^2 :



Bipolar doublets (+/- pair) at each edge. Zero-crossing marks exact edge location. Zero between doublets.

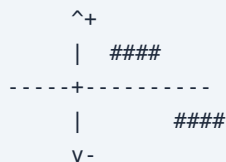
Positive spike at rising edge, negative spike at falling edge. Zero on flat plateau between.

Roof Edge (Triangular Profile)

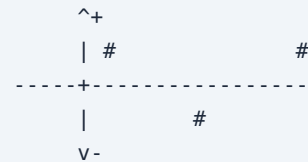
Intensity $f(x)$:



Gradient df/dx :



Laplacian d^2f/dx^2 :



Corrected: Positive impulses at two base corners; single negative impulse at apex. Linear ramps have zero second derivative — spikes at three corner discontinuities only. Formula: $\delta(x_0) - 2\delta(x_{\text{peak}}) + \delta(x_1)$.

Constant positive during rise, constant negative during fall. Zero at peak (sign change).

Parts B & C — Difference of Gaussians (DoG) Zero-Crossing Proof

The DoG is defined as the difference between two 2D Gaussians with standard deviations σ_1 and σ_2 ($\sigma_1 < \sigma_2$):

$$\text{DoG}(x,y) = G(x,y; \sigma_1) - G(x,y; \sigma_2)$$

$$\text{where } G(x,y; \sigma) = (1 / 2\pi\sigma^2) * \exp(-(x^2+y^2) / (2\sigma^2))$$

STEP-BY-STEP ZERO-CROSSING DERIVATION

Step 1 — Set DoG = 0 and equate the two Gaussians:

$$(1 / 2\pi\sigma_1^2) * \exp(-(x^2+y^2)/(2\sigma_1^2)) = (1 / 2\pi\sigma_2^2) * \exp(-(x^2+y^2)/(2\sigma_2^2))$$

Step 2 — Divide both sides (cancel $1/2\pi$) and take natural logarithm:

$$\ln(\sigma_2^2 / \sigma_1^2) = (x^2+y^2) * [1/(2\sigma_1^2) - 1/(2\sigma_2^2)]$$

Step 3 — Let $r^2 = x^2+y^2$ (radial distance). Solve for r^2 :

$$r^2 = x^2+y^2 = 2\sigma_1^2\sigma_2^2 / (\sigma_2^2 - \sigma_1^2) * \ln(\sigma_2^2/\sigma_1^2)$$

Step 4 — Express the zero-crossing variance σ^2 :

$$\sigma^2 = \sigma_1^2 * \sigma_2^2 / (\sigma_1^2 - \sigma_2^2) * \ln(\sigma_1^2/\sigma_2^2)$$

Step 5 — Substitute $k = \sigma_1/\sigma_2$ so $\sigma_1 = k\sigma_2$:

$$\sigma_1^2 = k^2 * \sigma_2^2 \Rightarrow \sigma^2 = k^2\sigma_2^2 * \ln(k^2) / (k^2 - 1)$$

$$\text{RESULT: } \sigma^2 = k^2 * \sigma_2^2 * \ln(k^2) / (k^2 - 1)$$

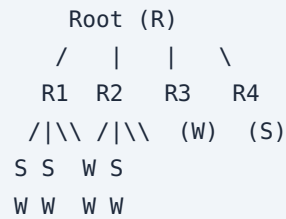
This gives the radial distance at which the DoG crosses zero — a ring-shaped zero-crossing contour approximating the Laplacian of Gaussian (LoG) edge detector.

Part D — Quad Tree Split & Merge

The 4x4 grid is divided into four 2x2 quadrants and analysed for homogeneity:

QUADRANT ANALYSIS

- **R1 (Top-Left):** Mixed → **SPLITS** into 4 pixels
- **R2 (Top-Right):** Mixed → **SPLITS** into 4 pixels
- **R3 (Bottom-Left):** All white → **DOES NOT SPLIT**
- **R4 (Bottom-Right):** All shaded → **DOES NOT SPLIT**



Legend: S=Shaded, W=White

R3, R4 are homogeneous leaf nodes.

R1, R2 split into 4 leaf pixels each.

MERGE PHASE

After splitting, adjacent leaf nodes may be merged if they satisfy the homogeneity criterion together. R3 (all white) and R4 (all shaded) are each homogeneous but not with each other — no merging occurs between them.

Chain Codes & GLCM

Part A — 12-Segment Boundary Description (Regular Octagon)

Clockwise traversal starting from top-right. Using 8-directional Freeman chain code:

Direction map (8-connectivity):

3	2	1
4	*	0
5	6	7

1. Chain Code (8-directional)

CC: 6 6 5 4 4 3 2 2 1 0 0 7

12 elements: 4 diagonal corners + 4 flat sides (2 steps each) for a regular octagon.

2. First Difference (rotation-invariant code)

First difference = $(CC[i] - CC[i-1]) \bmod 8$, where $CC[-1] = CC[\text{last}] = 7$

FD: 7 0 7 7 0 7 7 0 7 7 0 7

3. Shape Number (smallest integer rotation of First Difference)

SN: 0 7 7 0 7 7 0 7 7 0 7 7

Lexicographically smallest rotation — starts with 0, then repeating (7,7,0) pattern.

4. Signature Description

BOUNDARY SIGNATURE

The signature is the distance from a fixed interior reference point to the boundary as a function of angle. For this regular octagon, it produces a **periodic wave with 8 symmetric peaks** (corners) and **8 valleys** (flat sides). Period = $360/8 = 45$ degrees.

Part B — GLCM Construction

Image segment intensities: {1, 2, 3, 4, 5} → a **5x5 GLCM matrix**. Operator: 1 pixel right, 2 pixels above → displacement vector $d = (-2, +1)$, i.e., target = (row-2, col+1).

Valid Co-occurrence Pairs

#	Source (r,c)	Source val	Target (r-2, c+1)	Target val	Pair (i,j)
1	(2,0)	1	(0,1)	1	(1,1)
2	(2,1)	5	(0,2)	2	(5,2)
3	(2,2)	4	(0,3)	4	(4,4)

#	Source (r,c)	Source val	Target (r-2, c+1)	Target val	Pair (i,j)
4	(3,0)	5	(1,1)	1	(5,1)
5	(3,1)	4	(1,2)	2	(4,2)
6	(3,2)	5	(1,3)	2	(5,2)
7	(4,0)	2	(2,1)	5	(2,5)
8	(4,1)	3	(2,2)	4	(3,4)
9	(4,2)	1	(2,3)	3	(1,3)

Final 5x5 GLCM Matrix P

Row = source intensity i , Column = target intensity j . Total pairs = 9.

$i \setminus j$	1	2	3	4	5
1	1	0	1	0	0
2	0	0	0	0	1
3	0	0	0	1	0
4	0	1	0	1	0
5	1	2	0	0	0

$P(1,1)=1$, $P(1,3)=1$, $P(2,5)=1$, $P(3,4)=1$, $P(4,2)=1$, $P(4,4)=1$, $P(5,1)=1$, $P(5,2)=2$ |
Total = 9

GLCM INTERPRETATION

The GLCM encodes second-order texture statistics. Each entry $P(i,j)$ counts how often intensity i occurs at the source pixel when intensity j occurs at the displaced position. From this matrix, Haralick features such as **Energy**, **Contrast**, **Correlation**, and **Homogeneity** can be computed to classify image textures.